

Functional Connectivity for Motor Imaginary Recognition in Brain-computer Interface

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Abstract—Most brain-computer interfaces (BCIs) utilize univariate features (i.e., power spectrum or amplitude) for motor imagery (MI) pattern recognition, while less attention has been paid on multivariate analysis that considers information flow between various brain areas through brain connectivity estimations. Most recently, researches have proved that connectivity features were able to characterize different MI tasks. In this study, we investigated the performance of functional connectivity features measured by phase lag index (PLI), weighted phase lag index(wPLI) and phase-locking value(PLV) on MI classification. The widely-used filter-bank common spatial pattern (FBCSP) approach was employed here for performance assessment. A linear support vector machine was trained to classify different MI tasks using two publicly-available datasets from BCI Competition III and IV. The classification results showed that connectivity features achieved classification accuracy >85% in most cases and PLI outperformed all other methods including FBCSP. Our work suggested that functional connectivity features could be utilized as a powerful tool for recognizing different motor intention.

Index Terms—BCI, EEG, functional connectivity, phase lag index, weighted phase lag index, phase locking value

I. INTRODUCTION

Brain-computer interface (BCI) is a communication system that detects a subject's brain activities and translates them into control commands. Brain signals such as noninvasive electroencephalography (EEG) has been widely exploited as a measurement of user's intention in BCI. EEG-BCIs have been designed to assist neural function recovery for clinical purpose [1]. While previous work mainly focused on upper-limb application, more attention has been paid to identify lower-limb motor intention through motor imagery (MI) [2] [3]. However, most studies utilized the power or amplitude changes in some brain rhythms, such as event-related desynchronization (ERD) in alpha band and movement-related cortical potentials (MRCs) ([4] [5] [6]). Although promising results have been reported, these brain rhythms are easily contaminated by artifacts which caused sudden changes of amplitude. Moreover, as mentioned in [8]

power-based features have poor spatial resolution, making it difficult to distinguish left and right foot motor intention because the corresponding regions are too close [7].

It is widely accepted that different brain areas function together when dealing with sensorimotor and cognitive tasks. Such functional networks may provide more information and help better characterizing brain states and distinguishing various mental tasks. Researches have been made to explore connectivity features for motor imaginary recognition among which functional connectivity(FC) and effective connectivity have yield promising results [8]. Effective connectivity describes the direction and strength of information flow between different brain regions and reveals interactions of brain networks. It can be calculated by model-based method(i.e. dynamic causal modeling), data-driven method(i.e. Granger causality) and information theory-based method(i.e. Transfer entropy) [8] [9] [10], and have proved to be useful for MI task recognition [?] [12] [13]. Functional connectivity is defined as temporal correlation between separated brain regions which can be measured by linear, nonlinear and information-based methods [8] [9] [10]. Previous works have applied phase synchrony(PS) (e.g.instantaneous phase difference [14], synchrony rate [15]) to extract functional connectivity dynamics of brain states, and it was amplitude-free and thus being less sensitive to artifacts that cause abrupt power fluctuation.

In this paper we explored the performance of functional connectivity measured by phase-locking value (PLV), phase lag index (PLI) [16] and weighted phase lag index (wPLI) [17]. While previous applications attempted to capture short-time connectivity changes in certain frequency range during motor imagery [14] [15], we investigated FC features from different frequency bands of all pairs of electrodes and find the best combination for individual. A total frequency band ranged from 1-40Hz, and 3.5s EEG signals were used to calculate FC instead of short time values. The well-known filter-bank common spatial pattern (FBCSP) method was also applied for comparison.

The structure of this paper is as follows: Section II describes the EEG datasets used in this study and the calculation method. Section III presents and discusses the results. Section IV concludes this work.

II. METHOD AND MATERIAL

A. Public BCI Competition datasets

1) *BCI Competition III dataset IVa*: This dataset is provided by the Berlin BCI group, which includes five subjects (labeled as aa, al, av, aw, ay). During the experiment, subjects

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were asked to perform two types of MI tasks for 3.5s based on visual cues (1 for right hand and 2 for foot). Each type of motor imagery was performed 140 times, leading to 280 labeled trials per subject. EEG was recorded using BrainAmp amplifiers (Brain Products GmbH, Germany) and a 128 channel Ag/AgCl electrode cap from ECI. Specifically, 118-channel EEG was measured at positions of the extended international 10/20 system. Among these channels, we extracted 66 channels to reduce redundant information while keeping the main EEG characteristic (as shown in Fig. 1). Data was then down-sampled at 100Hz for following analysis. More details of this dataset could be found at <http://www.bbc.de/competition/iii/>

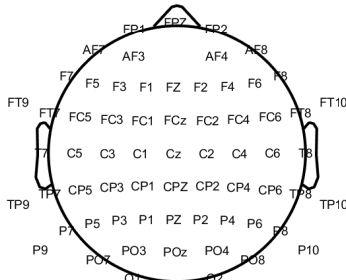


Fig. 1. Channel location selected for BCI Competition III dataset IVa

2) *BCI Competition IV dataset I*: This dataset is recorded from seven healthy subjects. Each subject performed two classes of MI tasks selected from three classes: left hand, right hand, and foot. EEG was recorded using BrainAmp amplifiers (Brain Products GmbH, Germany) and an Ag/AgCl electrode cap. The EEG signals were sampled at 1000 Hz and then down-sampled at 100 Hz from 59 electrodes. This study only used the calibration dataset. More details of the dataset could be found at <http://www.bbc.de/competition/iv/>

B. Connectivity Feature Calculation

In the current study, we employed three functional connectivity estimation methods, namely phase lag index (PLI), weighted phase lag index (wPLI), and phase-locking value (PLV). These connectivity values served as features for the following MI classification.

1) *Analytic Signal*: Wavelet transformation and Hilbert transformation can be implemented to get the instantaneous phase of signals. According to [18], they produce almost the same result. Here we use Hilbert transformation method. For a signal $x(t)$, the analytic form $X_a(t)$ is defined as:

$$X_a(t) = x(t) + i\bar{x}(t) = A(t)e^{i\phi(t)} \quad (1)$$

2) *Phase Lag Index*: PLI can be used to measure phase synchronization between two different electrodes while being less sensitive to volume conduction (VC) [16]. The idea is

that spurious connectivity induced by VC has phase difference around zero, while true connectivity causes constant positive or negative phase lag gap.

Let $X_l = A_l e^{i\phi_l}$ and $X_m = A_m e^{i\phi_m}$ be two analytical signals from two channels, and $S = X_l \times \text{conj}(X_m) = A_l A_m e^{i(\phi_l - \phi_m)}$, where $\text{conj}(X)$ is the conjugate of X . Then, the PLI between X_i and X_j could be estimated as:

$$PLI = \left| \frac{1}{N} \sum_{k=1}^N \text{sign}(\sin(\phi_l(k) - \phi_m(k))) \right| \in [0, 1] \quad (2)$$

where

$$\text{sign}(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } x < 0 \end{cases} \quad \phi_l(k) - \phi_m(k) \in (0, 2\pi] \quad (3)$$

Heuristically, larger PLI indicates stronger phase synchrony between two channels.

3) *Weighted Phase Lag Index*: Weighted PLI is an extension of PLI, which reduces the influence of phase jump caused by small perturbations [17]. Each PLI value is weighted by their absolute value of imaginary part. The wPLI between two signals (X_l and X_m) is calculated as follow:

$$wPLI = \left(\frac{\frac{1}{N} \sum_{k=1}^N [|Im(S_k)| \text{sign}(\phi_l(k) - \phi_m(k))]}{\frac{1}{N} \sum_{k=1}^N |Im(S_k)|} \right) \quad (4)$$

where $Im(S)$ is the imaginary part of S .

4) *Phase Locking Value*: Phase locking value between X_l and X_m is calculated as follow:

$$PLV = \left| \frac{1}{N} \sum_{k=1}^N e^{i(\phi_l(k) - \phi_m(k))} \right| \quad (5)$$

C. Evaluation scheme

EEG data were preprocessed using a standard pipeline, including re-reference by common average reference(CAR) approach, band-pass filtered using a six-order zero-phase ButterWorth filter. After that, nine frequency ranges (1–5, 5–9, 9–13, 13–17, 17–21, 21–25, 25–29, 29–33, and 33–37 Hz) of filtered signals were used for the following analysis. EEG from 0 to 3.5s post-stimulus of each frequency band was selected to calculate phase-based connectivity features. Note that we used 66 channels EEG, resulting in 2145 $(66 \times (66 - 1)) / 2$ features in each frequency band per trial. Thus we got $9 \times 3 \times 2145$ features for each subject. These features were selected by Fisher ratio that measured the discriminability of individual features [15]. Fisher ratio is calculated as:

$$F = \frac{(u_1 - u_2)^2}{v_1 + v_2} \quad (6)$$

where u_1 u_2 are means of class1 and class2, v_1 v_2 are variances of two classes. In this study, the top 400 features were selected to train a linear support vector machine (SVM) and classification accuracy was calculated using 10-fold cross-validation. The overall connectivity feature (take PLI as an example) calculation and selection algorithm is illustrated

in algorithm 1.

Algorithm 1 Phase Lag Index calculation and selection

Input Raw EEG $S \in R^{C \times T}$

1) Feature calculation

for $i = 1 : m$ **do**

1. band-pass filter in i^{th} frequency range
2. apply Hilbert transform to get instantaneous phase and amplitude
3. computing PLI for every pair of channels

end for

2) compute Fisher ratio for all features

3) choose features according to Fisher ratio

Output PLI features of MI tasks

D. CSP Features

CSP is a popular method for MI feature extraction and has been proved to achieve good results for binary classification problems. Heuristically, CSP finds spatial filters that maximize the variance of one class while minimizing the other. Many extensions of CSP have been developed, among which filter-bank CSP (FBCSP) provided satisfactory classification results for MI tasks. Therefore in the current study, we used FBCSP features as a reference to validate the performance of the proposed method. EEG data were band-pass filtered in thirteen frequency ranges (6–10, 8–12, ..., and 30–34 Hz) using the same filter as described above after CAR, and CSP features were calculated using EEG from 0 to 3.5s post-stimulus of each frequency band. Sparse

regression was implemented to select the most significant features as described in [19]. Four features were used to train a linear SVM for each subject, and 10-fold cross-validation was implied to get classification accuracy.

III. RESULTS AND DISCUSSIONS

Classification results for two datasets described above were depicted in Fig. 2. It could be seen that PLI features outperformed the other three methods in both datasets.

A. Classification results of BCI Competition III dataset IVa

Table 1 presents the classification results of BCI Competition III dataset. Specifically, functional connectivity features measured by three methods achieve high classification accuracy (>85%) for all subjects, where PLI achieved the highest accuracy for subject aa, al and av. In contrast, CSP got the highest accuracy for subject aw and ay. However, CSP yielded poor results for subject av, which is 18.21% lower than that of PLI, suggesting low generalizability of the CSP method. Of note, the classification accuracy by CSP in this study is consistent with previous results [19].

B. Classification results of BCI Competition IV dataset I

Table 2 presents the classification accuracy for seven subjects that perform different MI tasks in BCI Competition IV dataset. We found that PLI features achieved the highest average accuracy, which is 12.57% higher than that of CSP. PLI characteristics are comparatively better than CSP power features for subjects a, b, c, f, and g. In comparison with the CSP method, the classification accuracy improved by PLI for subjects a, b, c, f, and g is 16.5%, 35.5%, 24%, 11% and

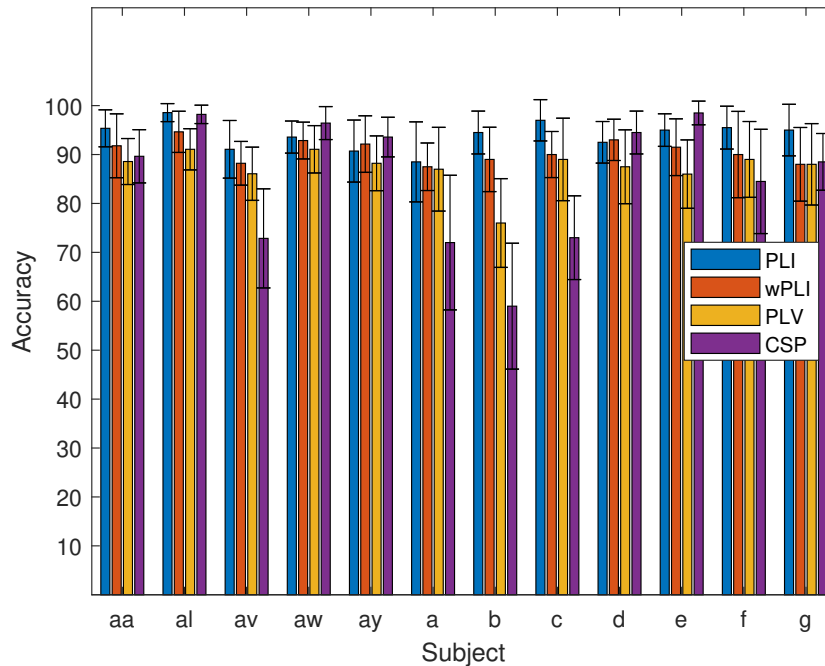


Fig. 2. Classification accuracy of all subjects from BCI Competition III dataset IVa and BCI Competition IV dataset I.

TABLE I
AVERAGE CLASSIFICATION ACCURACY AND STANDARD DEVIATION (%) FOR BCI COMPETITION III IVA DATASET

Subjects	PLI	wPLI	PLV	CSP
BCI Competition III IVa dataset				
aa	95.36 ± 3.78	91.79 ± 6.53	88.57 ± 4.70	89.64 ± 5.44
al	98.57 ± 1.84	94.64 ± 4.21	91.07 ± 4.21	98.21 ± 1.88
av	91.07 ± 5.89	88.21 ± 4.47	86.07 ± 5.44	72.86 ± 10.13
aw	93.57 ± 3.28	92.86 ± 3.76	91.07 ± 4.84	96.43 ± 3.37
ay	90.71 ± 6.34	92.14 ± 5.78	88.21 ± 5.60	93.57 ± 4.05
Average	93.86 ± 4.23	91.93 ± 4.97	89.00 ± 4.96	90.14 ± 1.32
BCI Competition III IVa dataset				
a	88.50 ± 8.18	87.50 ± 4.86	87.00 ± 8.56	72.00 ± 13.76
b	94.50 ± 4.38	89.00 ± 6.58	76.00 ± 9.07	59.00 ± 12.87
c	97.00 ± 4.22	90.00 ± 4.71	89.00 ± 8.43	73.00 ± 8.56
d	92.50 ± 4.25	93.00 ± 4.22	87.50 ± 7.55	94.50 ± 4.38
e	95.00 ± 3.33	91.50 ± 5.80	86.00 ± 6.99	98.50 ± 2.42
f	95.50 ± 4.38	90.00 ± 8.82	89.00 ± 7.75	84.50 ± 10.66
g	95.00 ± 5.27	88.00 ± 7.53	88.00 ± 8.32	88.50 ± 5.80
Average	94.00 ± 4.86	89.86 ± 6.07	86.07 ± 8.10	81.43 ± 8.35

7% respectively. For subjects d and e, PLI and CSP both got high accuracy (>90%) with CSP slightly outperformed PLI.

C. Discussion

Brain connectivity changes during motor imagine are not limited to specific brain regions and frequency bands, thus we explored connectivity features from wide frequency ranges and large scales in this study. Compared with single-channel power changes, connectivity features contain more information to characterize different brain states and can be useful for differing brain states. In the current exploratory study, functional connectivity features have proved to be efficient for MI pattern recognition, among which PLI features achieve the best results. Although CSP has been widely recognized as an effective method for binary MI task classification, it may perform poorly for some subjects (i.e. subject b of BCI Competition IV dataset 1). In contrast, PLI gets generally high accuracy for all the subjects, indicating that functional connectivity features can modulate different brain activities better than power features.

Since functional connectivity and band power revealed different characteristics of EEG signal, the parameters used for computing these features were inconsistent. For example, the number of connectivity features was set as 400 in this study while only 4 filter-bank CSP features were used. Besides, the frequency bands for the two methods were also different. Reasons for choosing these parameters are based on our preliminary study: 1) We first chose the number of FBCSP features according to [19]. To assess the influence of this parameter, we calculated the 10-fold cross-validation accuracy of different FBCSP feature numbers from 4 to 32 and it turned out that 4 features achieved the best performance in most cases, which is consistent with the results in [19]. The same procedure was used for connectivity feature number selection from 4 to 2000. Though the optimal

number was around 900, classification accuracy increased slightly as the feature number grew from 400 to 900, as such 400 features were used in current work; 2) In most cases, the selected FBCSP features were mainly within 6-30 Hz, which contains μ rhythm and β rhythm. However, connectivity features chosen by Fisher ratio were distributed in various frequency bands. Thus a wider frequency range was applied to extract connectivity characteristics.

It should be mentioned that Fisher ratio for connectivity features was calculated using the whole feature set, and then the result was used to select features and apply to 10-fold cross-validation. In the cross-validation procedure, the Fisher ratio method sometimes failed to extract discriminative characteristics when it was computed only using training data, while for other subjects features selected using training data showed promising results. Robust methods for feature selection need to further explore the practicability of functional connectivity-based feature construction approach for MI classification.

Some aspects of the study need to be taken into consideration when interpreting our findings. First, it is reported that functional connectivity should be measured on the EEG source space by source localization techniques [9], while this study focused on EEG sensor space, and the effect of volume conduction (VC) has not been fully considered although PLI method is insensitive to VC. Further work is expected to investigate the influence of VC and the performance of our methods at source space level. Second, the primary focus of the current study is to explore the feasibility of utilizing functional connectivity as features for MI classification. Although we have employed three widely-used EEG functional connectivity estimation methods (PLI, wPLI, and PLV) to construct the features, further studies are needed to assess and compare the performance of MI classification using

other connectivity estimation methods with directionality information (i.e., partial directed coherence, directed transfer function, etc.).

IV. CONCLUSION

EEG phase synchrony may contain more information than power features to distinguish various brain states. In this study, we demonstrated that functional connectivity changes during motor imagery could be useful for classification. Notably, phase lag index outperformed other connectivity calculation methods and yielded higher accuracy than CSP in most cases. Our result showed that functional connectivity features can achieve more reliable performance than CSP power features. Future work will further optimize the parameters to simplify procedure of connectivity calculation.

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